

Electrochemical Impedance Spectroscopy for Soft Biomaterials Formulation

Automated data processing and analysis workflow, towards self-driving labs for
materials discovery

Eindhoven University of Technology
Institute for Complex Molecular Systems

Enzo SIMIONI

Under the supervision of Doctor Annabel BASFORD
Within the group of Associate Professor Ghislaine VANTOMME

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Contents

Acknowledgements	i
Host Institution and Research Group	i
Nomenclature	ii
Abstract	iii
1 Introduction and context of this research	1
1.1 Motivation and background	1
1.2 Preliminary theory	2
2 Results and Discussion	3
2.1 Workflow	3
2.1.1 Equivalent circuit logic	3
2.1.2 Automated processing and validation	4
2.1.3 Analysis outputs and interpretation support	5
2.2 Experimental Results	6
2.2.1 Fixed UPy w/v%	6
2.2.2 Fixed PEDOT:PSS w/v%	7
2.3 Discussion	7
3 Materials and Methods	8
3.1 Materials	8
3.2 Experimental Methods	8
3.3 Computational methods	9
3.3.1 Automated outlier removal	9
3.3.2 Linear Kramers-Kronig validation	9
3.3.3 Fitting method	9
3.3.4 χ^2 and χ_{red}^2 parameters	10
3.3.5 Residual analysis	10
4 Conclusions and outlooks	11
4.1 Summary of the project	11
4.2 Limitations	11
4.3 Future Work	12
5 Appendices	13
References	18

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Host Institution and Research Group

This internship was carried out at Eindhoven University of Technology (TU/e), within the Institute for Complex Molecular Systems (ICMS) and the Supramolecular Chemistry and Materials research environment of the Vantomme group. ICMS is a multidisciplinary research institute that brings together chemistry, biology, physics, mathematics, and engineering to study and design complex molecular systems. Its research focuses on understanding how molecular structure, dynamics, and interactions give rise to functions at larger length scales, with applications ranging from advanced materials for photovoltaic cells to biomedical hydrogels [1, 2].

The Vantomme group focuses on the design, synthesis, and characterisation of complex molecular systems to create functional supramolecular materials. By combining synthetic and physical organic chemistry, the group aims to organise molecules into ordered and switchable structures, with long-term interests in adaptive materials, molecular computing, bio-(opto)electronics, and sustainable materials [3, 4]. The present work fits within this research direction by studying PEDOT:PSS–UPy conductive hydrogels, where supramolecular formulation controls both mechanical and electrical properties.

More specifically, this project contributes to the characterisation and analysis component of a future self-driving laboratory for supramolecular materials discovery, by converting EIS spectra into validated, reusable, and machine-readable descriptors.



Figure 1: (a) Group photo of the Vantomme group (b) TU/e diversity rainbow logo, Bart van Overbeeke
Image number 12832

Nomenclature

ω Angular frequency, $\omega = 2\pi f \text{ rad s}^{-1}$

AC Alternating Current

CE Counter Electrode

CP Conductive Polymer

CPE Constant Phase Element

DC Direct Current

DBSA 4-DodecylBenzeneSulfonic Acid

DL Double Layer

ECM Equivalent Circuit Model

EIS Electrochemical Impedance Spectroscopy

EP Electrode Polarisation

HCl Hydrochloric acid

HF High Frequency

HTE High-Throughput Experimentation

IQR Inter-Quartile Range

LF Low Frequency

lin-KK Linear Kramers-Kronig

NaOH Sodium Hydroxide

OMIEC Organic Mixed Ionic-Electronic Conductor

PBS Phosphate-Buffered Saline

PEDOT Poly(3,4-ethylenedioxythiophene)

PEDOT:PSS Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate)

PSS Poly(styrenesulfonate)

RE Reference Electrode

WE Working Electrode

Abstract

English

Conductive hydrogels are promising materials for soft bioelectronic interfaces as they combine a hydrated, tissue-like mechanical environment with the possibility of both ionic and electronic charge transport. In this context, PEDOT:PSS hydrogels reinforced by a secondary non-covalent UPy network are attractive candidates, as PEDOT:PSS aggregates can provide mixed conduction while the supramolecular network can improve softness. However, both the mechanical and electrical properties of these materials depend strongly on formulation, including the relative amount of PEDOT:PSS, UPy, water, and mobile ions, meaning that their behaviour must be measured systematically rather than inferred from composition alone. Although electrochemical impedance spectroscopy (EIS) is often associated with battery and electrolyte studies, its ability to probe a material's response over a broad frequency range makes it particularly relevant for biological applications, where stimulation and sensing occur under alternating and time-dependent electrical conditions. However, EIS interpretation is often cumbersome, subjective, and difficult to scale. This becomes particularly important in the new age of materials discovery, where high-throughput characterisation methods play an increasing role. This report therefore presents an object-oriented Python workflow for automated and tunable EIS preprocessing, validation, fitting, analysis, and plotting. The results show that automated EIS analysis can accelerate the identification of conductivity trends in PEDOT:PSS, UPy-reinforced hydrogels while providing both human and machine-readable results and validation metrics.

Français

Les hydrogels conducteurs constituent des matériaux prometteurs pour faire office d'interfaces bioélectroniques souples. Ces gels combinent en effet un environnement mécanique hydraté, proche de celui des tissus biologiques, avec la possibilité d'assurer un transport de charges tant ionique qu'électronique. Dans ce contexte, les hydrogels de PEDOT:PSS renforcés par un réseau secondaire non covalent de motifs UPy sont des candidats intéressants. Les agrégats de PEDOT:PSS peuvent assurer une conduction mixte, tandis que le réseau supramoléculaire à base d'UPy améliore la souplesse du matériau. Cependant, les propriétés mécaniques et électriques de ces matériaux dépendent fortement de la formulation, notamment des proportions relatives de PEDOT:PSS, d'UPy, d'eau et d'ions mobiles. Leur comportement doit donc être mesuré de manière systématique, à défaut de pouvoir être déduit de la seule composition. Bien que la spectroscopie d'impédance électrochimique (EIS) soit souvent associée à l'étude des batteries et des électrolytes, sa capacité à sonder la réponse d'un matériau sur une large gamme de fréquences la rend particulièrement pertinente pour les applications biologiques, où la stimulation et la détection s'effectuent dans des conditions électriques alternatives et dépendantes du temps. Toutefois, l'interprétation des données d'EIS est souvent fastidieuse, subjective et difficile à transposer à grande échelle. Cet aspect devient particulièrement important dans le contexte actuel de la découverte de matériaux, où les méthodes de caractérisation à haut débit jouent un rôle croissant. Ce rapport présente donc une architecture Python orientée objet, automatisée et modulable, pour le prétraitement, la validation, l'ajustement, l'analyse et la représentation graphique de données d'EIS. Les résultats montrent que l'analyse automatisée des données d'EIS peut accélérer l'identification de tendances de conductivité dans des hydrogels, tout en fournissant des résultats et des métriques de validation lisibles à la fois par l'utilisateur et par la machine.

1 Introduction and context of this research

1.1 Motivation and background

According to the World Health Organization, more than 15 million people live with spinal cord injuries worldwide [5]. Although surgery and rehabilitation strategies can improve daily life, no curative treatment is currently available. This is largely linked to the limited regenerative capacity of the adult central nervous system. A major challenge is that functional recovery requires damaged axons not only to regrow, but also to re-establish appropriate biochemical and electrical communication within the spinal neural network. In order to provide an adequate regenerative environment, tissue-like mechanical softness is required, making conductive hydrogels attractive candidates [6]. Their polymer phase provides structural integrity, and when conductive polymers are included, electronic transport pathways. While their liquid phase enables swelling, diffusion, nutrient transport, and ion mobility [7, 8]. Due to their high water content and low elastic modulus, hydrogels can better approximate soft biological environments than rigid electronic materials [9].

Among conductive hydrogel systems, PEDOT:PSS (Figure 1.1b) is particularly relevant, since its conductive function is based on both electronic and ionic pathways. Poly(3,4-ethylenedioxythiophene) (PEDOT) provides the oxidised, hole-conducting conjugated phase, whereas poly(styrene sulfonate) (PSS) acts as a charge-balancing, hydrophilic polyelectrolyte phase. Electronic transport occurs mainly through connected PEDOT-rich domains [7, 10], while ionic transport is enabled by hydrated PSS-rich regions and the surrounding aqueous phase. The materials studied in this project combine PEDOT:PSS with monofunctional (M-UPy) and bifunctional (B-UPy) ureido-pyrimidinone (UPy) derivatives (Figure 5.1). UPy motifs are self-complementary units that dimerise through reversible quadruple hydrogen bonding. In this system, B-UPy acts as a crosslinker that connects UPy associations into a dynamic supramolecular network, while M-UPy helps tune the density and homogeneity of these associations. The mixture therefore gives a more controllable secondary network than B-UPy alone, helping balance cohesion, softness, and conductivity [11, 12]. However, introducing this non-conductive secondary network may also disturb the connectivity of PEDOT-rich domains or ionic mobility. The formulation problem is therefore a trade-off between softness and conductivity.

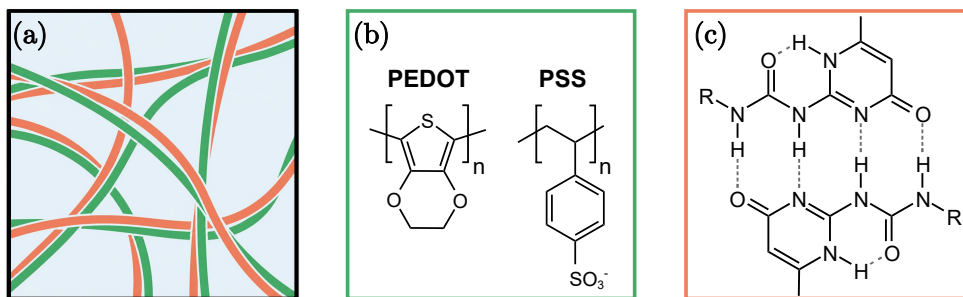


Figure 1.1: (a) Schematic view of PEDOT:PSS conductive pathways (green) and the UPy secondary network (orange), inspired by [13]. (b) Molecular structures of PEDOT and PSS polymers. (c) General molecular structure of crosslinked UPy motifs, with hydrogen bonds in grey. Corresponding R groups are given in the appendix (Figure 5.1)

Since the conductivity involved in this balance arises from both ionic and electronic transport, a single direct-current conductivity value obtained by four-point probe measurement does not fully capture the different contributions within the system. Bioelectronic materials operate under time-dependent conditions involving ionic migration, capacitive charging, interfacial polarisation, and distributed relaxation pro-

cesses [14]. Electrochemical impedance spectroscopy (EIS) is therefore more informative in this context due to its ability to probe complex impedance $\underline{Z}$ over a broad range of frequencies [15, 16]. This allows for the decoupling of the different conductive contributions and capacitive behaviours, which is particularly important for conductive hydrogels [17, 18]. EIS is thus used here not only to estimate conductivity, but also to evaluate how PEDOT:PSS loading and the UPy network influence the frequency-dependent electrical behaviour of the hydrogel.

Although EIS is a well-established technique, its application to soft mixed ionic-electronic hydrogels remains challenging due to the complex nature of the materials. Furthermore, materials discovery increasingly relies on screening multi-dimensional formulation spaces, where automated and reproducible characterisation and analysis become essential as material synthesis throughput increases [19]. Existing EIS workflow tools are predominantly in the field of battery electrolytes [19]. This project thus provides an automated workflow, tailored to complex hydrogel systems to support high-throughput EIS analysis. The workflow remains open-source, modifiable, and still suitable for smaller studies.

1.2 Preliminary theory

The interpretation of EIS is only meaningful if the measured system satisfies a set of basic assumptions. The response should be linear with respect to the applied perturbation, causal, and time-invariant over the duration of the frequency sweep [15, 16]. These requirements are particularly important for hydrogels because water evaporation, swelling, ion redistribution, electrode polarisation, and contact changes can modify the measured impedance during acquisition. In this work, these assumptions are addressed experimentally by using small perturbation amplitudes (see appendix Table 5.2), reproducible measurement conditions, and computationally by inspecting residuals, fitting errors, and linear Kramers-Kronig validation (see Section 3.3).

In a voltage-controlled EIS experiment, the sample is perturbed with a sinusoid of small amplitude E_0 ,

$$\text{Applied: } E(t) = E_0 \sin(\omega t), \quad \text{Measured: } I(t) = I_0 \sin(\omega t + \phi).$$

These measurements are repeated for several angular frequencies $\omega = 2\pi f$ and the complex impedance is then:

$$\underline{Z}(\omega) = \frac{\mathcal{F}(E(t))}{\mathcal{F}(I(t))} = \frac{\underline{E}(\omega)}{\underline{I}(\omega)} = Z'(\omega) + jZ''(\omega) \quad (1.1)$$

where Z' and Z'' are the real and imaginary components of the impedance and $\mathcal{F}$ the Fourier transform operator. An EIS measurement is thus carried out on a certain frequency range and frequency increment fixed by the operator. Each frequency giving two measured values: Z' and Z'' . They can either be plotted one against the other in a Nyquist plot (appendix Figure 5.2a). Or, by using the exponential defined in Equation (1.2), the modulus $|Z|$ and the phase φ (appendix Figure 5.2b, c) of the complex impedance can be plotted against the frequency range in Bode plots.

$$\underline{Z}(\omega) = |Z(\omega)| \times e^{j\varphi(\omega)} \quad (1.2)$$

The measured spectra are then fitted with equivalent circuits (Section 2.1.1) in order to separate overlapping resistive, capacitive, ionic, electronic, and interfacial contributions. The workflow therefore makes it possible to compare formulation-dependent electrical responses in a more systematic way, while keeping the fitting and validation steps traceable. Although the PEDOT:PSS–UPy series are still prepared and measured manually in this work, the automated analysis represents a first step towards a closed-loop, automated workflow, linking formulation, characterisation, and machine learning optimisation (Figure 4.2).

2 Results and Discussion

2.1 Workflow

2.1.1 Equivalent circuit logic

In order to decouple the different aforementioned contributions to the impedance response, the experimental spectra were fitted with equivalent-circuit models (ECMs). The fitting process described in [Section 3.3.3](#) translates the frequency-dependent impedance of the material into a compact set of fitted electrical elements. The usual strategy is to start from the physical processes expected in the system, propose the simplest circuit able to represent them, and refine it using literature, experiments, and fit diagnostics ([Section 3.3](#)). The choice of ECM is therefore not only a mathematical fitting choice, but also a physical hypothesis about the dominant processes occurring in the sample [\[15, 16\]](#).

For PEDOT:PSS hydrogels, the starting hypothesis is the coexistence of ionic and electronic pathways [\[7, 8\]](#). Ion migration through the hydrated phase is represented by R_i , in series with a capacitor, modelling electrode polarisation. In parallel, PEDOT-rich domains can support electronic transport through polaronic and bipolaronic charge carriers along the oxidised PEDOT conjugated chains [\[20\]](#). This electronic pathway is represented by R_e and the resulting circuit is shown in [Figure 2.1a](#) with an example of the frequency regions where each component's contribution dominates ([Figure 2.1b](#)).

However, to refine the model, the interfacial polarisation element is not actually described with an ideal capacitor, but rather with a constant phase element (CPE), the impedance of which are defined as follows:

$$\text{Ideal: } Z_{\text{Capacitor}}(\omega) = \frac{1}{Cj\omega} \quad \text{Non-ideal: } Z_{\text{CPE}}(\omega) = \frac{1}{Q(j\omega)^\alpha} \quad (2.1)$$

where C (F) is a capacitance, Q ($F \cdot s^{\alpha-1}$) controls the magnitude of the pseudo-capacitive response, while α describes the deviation from ideal capacitive behaviour [\[15, 17\]](#) by taking values between 0 and 1 (1 being ideal). This replacement is motivated by the complex and heterogeneous nature of hydrogels, which often leads to non-ideal impedance responses, visible for instance in [Figure 5.2a](#), where the Nyquist plot shows a slightly depressed semicircle followed by a low-frequency (right part) diffusion-related tail.

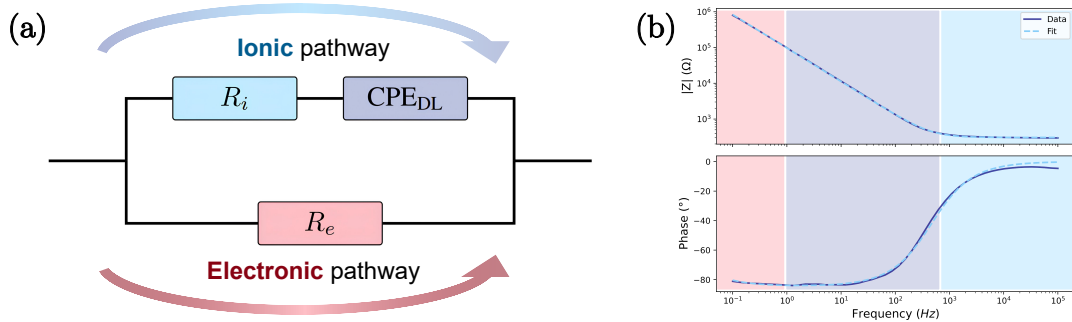


Figure 2.1: Equivalent-circuit logic used to separate dominant ionic and electronic contributions. (a) Schematic ECM in which R_i represents ionic resistance, CPE_{DL} the non-ideal double-layer capacitive response, and R_e the electronic resistance [\[18\]](#) (b) Colour-coded Bode plots illustrating the frequency regions where different circuit contributions dominate. Figure adapted conceptually from [\[15\]](#), see [Equation \(5.1\)](#) in the appendix for additional remark.

The chosen circuit topology can be fully described by its model impedance $Z_{\text{ECM}}(\omega, \mathbf{\Pi})$, where $\mathbf{\Pi}$ is the vector of fitted circuit parameters defined in [Equation \(2.2\)](#) in the case of [Figure 2.1a](#)'s circuit. The

fitting process described in Section 3.3 then allows for the extraction of fitted parameters $\mathbf{\Pi}_{\text{fit}}$. These parameters are then analysed, for example to identify conductivity trends (Section 2.1.3).

$$\mathbf{\Pi} = (R_i, Q_{\text{DL}}, \alpha_{\text{DL}}, R_e) \quad (2.2)$$

Remark: To avoid ambiguity in the notations, real scalar quantities are written in standard italic form, for example Z' and Z'' . Vectors or multi-component quantities are written in bold, such as the circuit parameters vector $\mathbf{\Pi}$ (see Section 3.3.3). Complex quantities, like $\underline{Z}(\omega)$, are underlined to indicate that they contain both real and imaginary components.

2.1.2 Automated processing and validation

The main production of this project is an automated and configurable workflow for EIS analysis. Instead of using a single monolithic script or a Jupyter notebook [6, 18], the code was divided into dedicated classes for configuration loading, preprocessing, fitting, validation, analysis, and plotting. This structure is preferable as each of these classes can be tested, modified, reused, and applied consistently across many experiments, improving maintainability, reproducibility, and scalability compared with a single all-in-one script. This approach also separates scientific choices from implementation details: the user defines circuits, initial guesses (see next paragraph), frequency windows, validation rules, conductivity mappings, plotting limits, and output preferences in root-level configuration files, separated from the rest of the codebase.

The usual workflow proceeds as follows: researchers start by defining their ECMs to be tried and provide for each component a set of initial guesses $\mathbf{\Pi}_{\text{init}}$ for the fitted parameters before optimisation (Section 3.3.3). This is necessary because ECM fitting is a nonlinear inverse problem: different starting values can lead to different local minima or even fail to converge [15, 21]. In practice, initial guesses were either chosen from literature or adjusted manually via trial and error. Users then have to provide their fitting, validation, analysis and plotting preferences (more details in the documentation [22]). Once that has been set up, the script can be executed and the pipeline (Figure 2.2) will access all the necessary classes in the project to process any new experimental data that has been added to the data folder (appendix Figure 5.3b). The present subsection focuses on the structure and traceability of the automated workflow; the analysis outputs generated by this pipeline are discussed in Section 2.1.3.

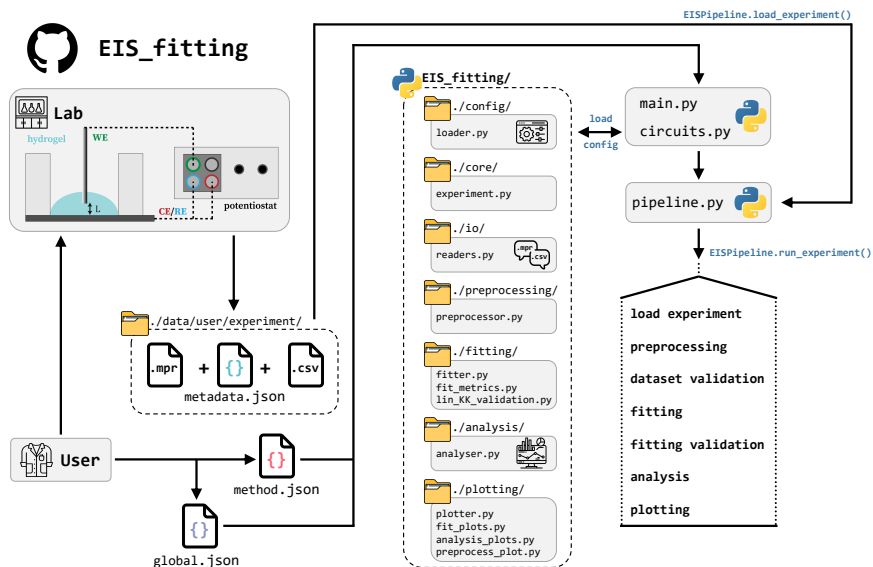


Figure 2.2: Schematic of the automated EIS processing and analysis pipeline [22]

The main advantage over manual fitting is reproducibility. Each spectrum is processed with the same frequency range, outlier rule, Lin-KK settings, weighting method, parameter-error threshold, and plotting style (Section 3.3) and all these parameters are stored to ensure optimal repeatability. This workflow is also more adaptable than built-in potentiostat software: project-specific circuits can be added in a specific file, with element names and colours mapped to physical meanings, conductivities calculated only from resistances explicitly assigned to through-sample transport (user-specified in mapping). This last part is essential for hydrogels as contact resistance, electrode polarisation, and interfacial processes may also appear as resistive elements but should not automatically be converted into material conductivity without physical justification [15, 18, 23].

2.1.3 Analysis outputs and interpretation support

The first output of the pipeline is the set of fit-quality and reliability metrics associated with each fitted spectrum. The workflow stores χ^2 , χ_{red}^2 , residuals, Lin-KK information, fitted-parameter standard deviations, and relative parameter errors (Section 3.3). These quantities are not reinterpreted here, since their definitions are given in Section 3.3, but they are essential for deciding whether a parameter trend is physically meaningful or not. Fitted values are only considered reliable when associated with small and random residuals, successful Lin-KK validation and acceptable parameter uncertainty [21, 24, 25].

For successful fits, the analysis stage computes conductivities for all resistances mapped as a through-sample transport resistance, using the sample geometry defined in the `metadata.json` (see appendix Figure 5.3 for more details on the file structure) of the experiment using:

$$\sigma = \frac{L}{RA} \quad (2.3)$$

where L is the sample thickness and A the electrode area. This conversion assumes that the selected resistance R represents a through-sample transport pathway in a well-defined measurement geometry; otherwise, the calculated value should only be treated as an apparent conductivity, since the fitted resistance may also contain contact, electrode-polarisation, or interfacial contributions [15]. Although some studies compare conductivity values extracted from EIS fitting with bulk conductivity obtained from DC measurements [6], this comparison should be made with caution. The EIS-derived value depends on the selected ECM, the assigned resistance element, the electrode geometry, and possible interfacial contributions. For this reason, the workflow includes an option to normalise conductivities by the minimum value in a series. Such normalisation does not replace absolute conductivity reporting, but it helps highlight relative trends between formulations while reducing the risk of overinterpreting an apparent conductivity as a true bulk property.

Finally, the pipeline produces human-interpretable and publication-ready figures of the aforementioned outputs (appendix Figure 5.3c). Fit plots (Figure 5.5a,b) show the agreement between the experimental and fitted spectra in Bode and Nyquist representations, and residual plots (Figure 5.5c) reveal frequency-dependent misfit. Parameter bar plots compare the mapped circuit elements (Figure 5.5d), and the conductivity summaries highlight the transport trends most relevant to the material design (Figure 2.4). This automated plotting is not only a convenience step: it ensures that all samples are displayed with consistent scales, colours, labels, and output formats, which improves reproducibility and interpretability while also making comparisons less dependent on manual formatting. In this sense, the workflow follows the broader goal of automated electrochemical data analysis: to make high-throughput or formulation-screening experiments easier to interpret and compare [19, 22].

2.2 Experimental Results

Following the experimental and computational workflow described in Chapter 3, the EIS spectra of the PEDOT:PSS–UPy hydrogels were fitted with different ECMs in order to obtain satisfactory fits. The extracted parameters allow for the study of the two series of samples measured, either fixing UPy (Section 2.2.1) or PEDOT:PSS (Section 2.2.2) w/v%.

2.2.1 Fixed UPy w/v%

The first gel-state series consisted of PEDOT:PSS–UPy hydrogels with a fixed UPy content of 1.2 w/v% and increasing PEDOT:PSS loading. This series directly tests the hypothesis introduced in Section 1.1: increasing PEDOT:PSS should improve electronic-network connectivity. The expected trend is a decrease in fitted through-sample resistance with increasing PEDOT:PSS content, corresponding to an increase in conductivity (Equation (2.3)).

The circuit used for the fitting of this dataset (Figure 2.3) was adapted from the PEDOT:PSS conductive hydrogel model reported by Feig et al. [6]. The model separates a cell or contact resistance R_c , an ionic branch R_i – CPE_{DL} , an electronic resistance R_e , and a gel pseudo-capacitive contribution CPE_g . This topology is consistent with the mixed ionic–electronic nature of PEDOT:PSS hydrogels while also encompassing other physical phenomena.

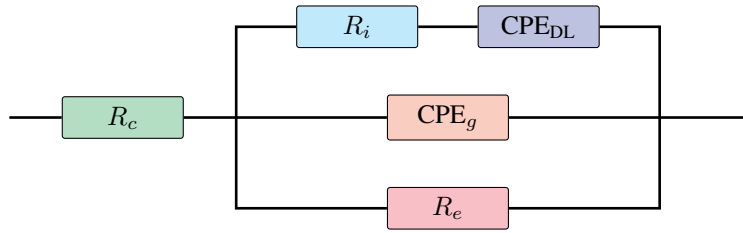


Figure 2.3: Equivalent circuit model `circuit_2`, shown with its colour and name mapping used in Figure 5.6. This circuit topology has previously been used to fit EIS data from PEDOT:PSS-based conductive hydrogels [6, 18].

The fitted conductivity trends extracted from the fitting with `circuit_2` are shown in Figure 2.4. The normalised electronic conductivity increases strongly with PEDOT:PSS loading, which is consistent with the expected improvement of PEDOT-rich network connectivity. The ionic conductivity also increases from P0.1U1.2 to P1U1.2, but then decreases slightly for P2U1.2. This suggests that the ionic contribution does not simply scale with PEDOT:PSS content: at high PEDOT:PSS loading, changes in gel morphology, water distribution, or PSS-rich domains may affect ion mobility. Since the conductivities are normalised, these results should be interpreted mainly as relative formulation-dependent trends rather than absolute bulk conductivities.

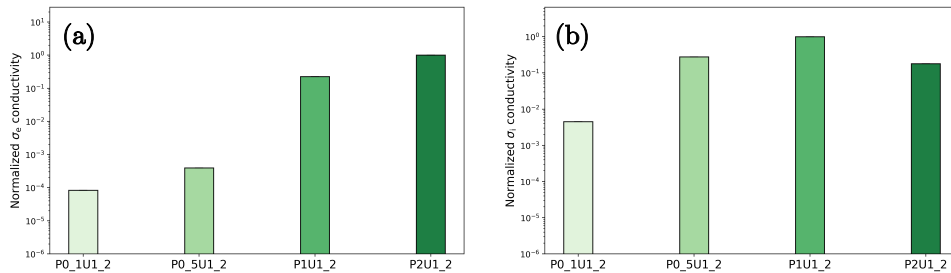


Figure 2.4: Normalised conductivities (electronic σ_e (a) and ionic σ_i (b)) respectively extracted from the resistance values fitted with `circuit_2` (Figure 2.3) R_e and R_i .

2.2.2 Fixed PEDOT:PSS w/v%

The second gel-state series consisted of PEDOT:PSS–UPy hydrogels with fixed PEDOT:PSS content and varying UPy loading. This series probes whether increasing the secondary supramolecular network modifies the impedance response and the fitted transport descriptors. Because the spectra could not be described as well by `circuit_2`, the more flexible `circuit_3` (Figure 5.7) was used. This model led to successful fits using two successive R –CPE relaxation blocks and a terminal CPE associated with interfacial polarisation. However, at this stage these two fitted resistive contributions cannot be uniquely assigned to ionic and electronic transport. They are therefore interpreted as effective conductivity descriptors rather than as separated physical pathways. More details on the circuit and fitted parameters are provided in the [Appendices](#).

The normalised conductivities extracted from `circuit_3` are shown in Figure 2.5. The first conductivity descriptor, σ_1 , increases with UPy content, reaching its highest value for P2U2.5. This indicates that the resistance associated with the high-frequency R –CPE block (see [Appendices](#)) decreases as the supramolecular network content increases.

The second descriptor, σ_2 , shows a different trend. It increases from P2U0.6 to P2U1.2, but decreases again for P2U2.5, while remaining higher than for P2U0.6. This suggests that the contribution represented by the second R –CPE block may be favoured at intermediate UPy content, but is not improved monotonically by further increasing the supramolecular network. A possible interpretation is that moderate UPy loading improves gel continuity, hydration, or structural stability, whereas higher UPy content may begin to modify ion mobility, phase organisation, or the connectivity of conductive PEDOT:PSS-rich domains.

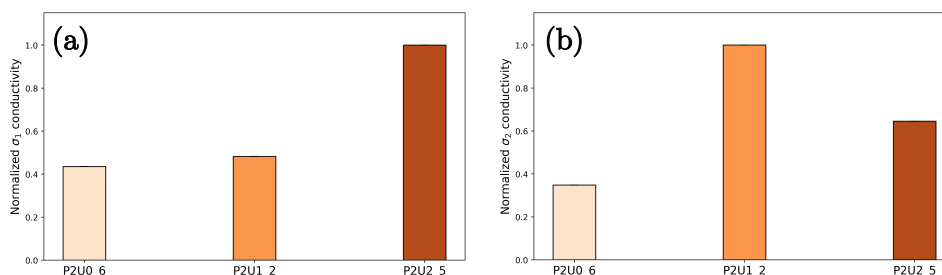


Figure 2.5: Normalised conductivities extracted from the fitted resistance elements of `circuit_3` for the fixed PEDOT:PSS series with varying UPy content. Panels (a) and (b) correspond respectively to the normalised conductivities σ_1 and σ_2 extracted from R_1 and R_2 (Section 2.1.3)

2.3 Discussion

The results of this work highlight the value of an automated and validation-aware EIS workflow for conductive hydrogel development. Each sample is linked to its raw and cleaned spectra, validation metrics, fitted parameters, parameter uncertainties, conductivities, and plots. This improves reproducibility and traceability while keeping the workflow scalable to larger datasets.

Experimentally, the objective is not simply to maximise PEDOT:PSS content, but to balance conductivity, softness, processability, stability, and biological compatibility. The fixed-UPy series shows that increasing PEDOT:PSS improves the fitted conductive response, while the fixed-PEDOT:PSS series suggests that UPy content also influences the impedance behaviour. Since the U1.2 formulation also exhibits an optimal stiffness for tissue-like applications [13] with a low Young’s modulus close to 1 kPa [26], the best hand-selected compromise identified in this study is P2U1.2. In future work, this type of formulation optimisation will be made through a closed-loop, self-driving materials-discovery workflow rather than by manual screening (Section 4.3).

3 Materials and Methods

3.1 Materials

Materials: Monofunctional UPy (UPy-OEG₁₂-Gly) (M-UPy, Figure 5.1a) and Bifunctional UPy (BF UPy-PEG_{10k}-UPy) (B-UPy, Figure 5.1b) reagents were purchased from SyMo-Chem. PEDOT:PSS was purchased from Sigma-Aldrich (900208-1G). All chemicals were used as received unless otherwise stated.

Preparation: B-UPy was dissolved in PBS and heated to 80°C for 10 minutes without stirring. M-UPy underwent the same procedure in PBS containing 80 mM NaOH. Each sample was then briefly vortexed and allowed to cool to room temperature. M-UPy was then neutralised with 1 M aqueous HCl. PEDOT:PSS was dissolved by stirring for 10 minutes in Milli-Q water. The resulting solution was then used to mix with the M-UPy and B-UPy solutions (1:1 M-UPy:B-UPy molar ratio) at varying weight/volume % of PEDOT:PSS/UPy. The resulting hydrogels were then equilibrated at room temperature overnight to form fibres. From this methodology, the following nomenclature was used: **P_xU_y** where **P** = PEDOT:PSS, **U** = UPy and *x* and *y* denote their weight percentage (w/v%), respectively.

3.2 Experimental Methods

EIS measurements presented in this report were performed with a BioLogic VSP single-channel, three-connections potentiostat used in a two-electrode configuration shown in Figure 3.1. The counter electrode (CE) and reference electrode (RE) leads were both connected to the same bottom platinum plate electrode. The working electrode (WE) connection was attached to a platinum electrode, inserted at the top of the well. Samples of 250 µL were manually transferred into the well for manual measurements.

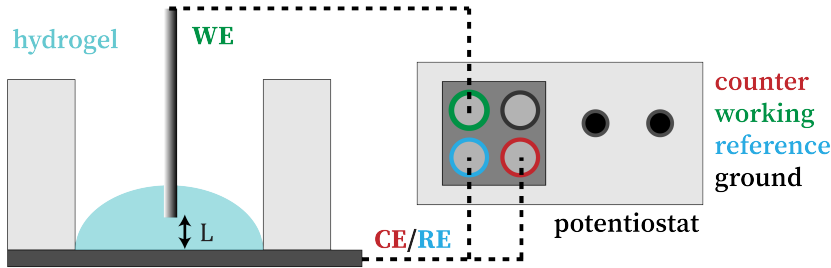


Figure 3.1: Two-electrode setup for EIS measurements on gel-state hydrogels, platinum bottom plate (CE/RE combined, in dark grey) and platinum rod electrode (WE). See Table 5.3 for measurements.

Remark: The two-electrode configuration is widely used for EIS measurements on conductive hydrogels [6, 18, 23] rather than three-electrode configuration. This geometry is particularly relevant for bulk gel studies as it probes a device-like current path through the material. However, the measured impedance also includes electrode polarisation and contact contributions, which must therefore be considered when interpreting the fitted circuit parameters [18].

To reduce misinterpretation, the experimental workflow included repeated frequency sweeps (three per sample). Repeated sweeps make it possible to test whether the spectrum changes with time, which is a practical test of stability and time invariance (see Section 1.2).

3.3 Computational methods

3.3.1 Automated outlier removal

The automated workflow can optionally remove isolated outliers before fitting using a quantile/IQR rule applied to selected spectral features, such as $\log_{10} |Z|$ and phase φ [27]. In the current implementation, the quartiles are computed over the whole selected frequency range for each feature. A point measured at frequency ω is removed if at least one selected feature X satisfies:

$$X(\omega) \notin [Q_1(X) - t \times \text{IQR}(X), Q_3(X) + t \times \text{IQR}(X)] \quad \text{IQR}(X) = Q_3(X) - Q_1(X) \quad (3.1)$$

where $Q_1(X)$ and $Q_3(X)$ are the first and third quartiles of that feature X and t is the user-defined threshold in `global.json` [22] (Figure 5.4). Since impedance magnitude and phase vary systematically with frequency, this global IQR criterion is not intended to define the physically relevant frequency range. It is only used as a conservative filter for isolated corrupted points. For this reason, the option can be disabled and the filtered features and threshold can be adjusted. Moreover, the number of removed points is recorded in the output CSV for traceability and printed in the terminal.

3.3.2 Linear Kramers-Kronig validation

Upon removal of the outliers, linearity of the datasets was assessed using the linear Kramers-Kronig (Lin-KK) procedure. In this method, measured impedance $Z_{\text{exp}}(\omega)$ is approximated by a series resistance R_Ω and a set of relaxation elements (RC) (circuit given in appendix Figure 5.9)

$$Z_{\text{Lin-KK}}(\omega) = R_\Omega + \sum_{k=1}^M \frac{R_k}{1 + j\omega\tau_k} \quad \text{where} \quad \tau_k = R_k C_k \quad (3.2)$$

This auxiliary circuit is not interpreted as the physical model of the sample; it is used to test whether the spectrum is compatible with the assumptions of linear, causal, and stable EIS data [25, 28]. If Lin-KK validation fails and the `reject_on_fail` option is enabled in `global.json` (Figure 5.4), the spectrum is rejected before the fitted ECM parameters are interpreted.

3.3.3 Fitting method

Equivalent-circuit fitting was performed by complex non-linear least squares optimization using the fitting function of the `impedance.py` [29] Python library. The fitting algorithm uses the set of parameters used to describe the ECM, $\mathbf{\Pi} = (R_e, R_i, Q_{\text{DL}}, \alpha_{\text{DL}}, \dots)$ defined in Equation (2.2), and minimizes the individual distances $|Z_{\text{exp}}(\omega) - Z_{\text{fit}}(\omega, \mathbf{\Pi})|$ for all ω , starting from a set of initial guesses $\mathbf{\Pi}_{\text{init}}$ (see Section 2.1.2). The goal of the fitting can be summarized as minimising the objective function χ^2 defined in Equation (3.3). Since the problem is non-linear, convergence to the global minimum is not guaranteed and depends on the ECM and initial guesses (Section 2.1.2) [15].

$$\chi^2(\mathbf{\Pi}) = \sum_{\omega} \left| \frac{Z_{\text{exp}}(\omega) - Z_{\text{fit}}(\omega, \mathbf{\Pi})}{Z_{\text{exp}}(\omega)} \right|^2 \quad (3.3)$$

Remark: In the present workflow, modulus weighting (division by $Z_{\text{exp}}(\omega)$) was used so that high-impedance regions did not dominate the objective function because of their magnitude [24].

The fitted circuit parameters obtained ($\mathbf{\Pi}_{fit}$) were then saved with their standard deviations to a circuit-specific CSV file using human-readable names defined in the circuit mapping. If a fitted parameter had a percentage error above the threshold defined by `validation.error_percent_param_threshold` parameter in `global.json` (set to 100% in Figure 5.4), the fit was rejected. This rule prevents visually acceptable but non-reproducible fits from being interpreted as physically meaningful.

3.3.4 χ^2 and χ_{red}^2 parameters

After fitting, the workflow stores the minimised objective function $\chi^2(\mathbf{\Pi}_{min}) = \chi^2$, as `chi_square` in the parameter CSV files. This value can also be displayed on the Bode, Nyquist, and residual plots to facilitate comparison between fits. However, the raw value of χ^2 is not always sufficient for comparing different candidate circuits. A more complex circuit contains more adjustable parameters in $\mathbf{\Pi}$, hence giving greater mathematical flexibility and resulting in lower χ^2 , even if the additional elements are not physically justified [15, 21]. It is therefore useful to normalise the fitting error by the number of degrees of freedom in the dataset [16, 24] $\nu = 2N - p$, where N is the number of valid frequencies after outliers removal, and p the number of fitted parameters in the circuit. χ_{red}^2 is hence defined as follows:

$$\chi_{red}^2 = \frac{\chi^2}{\nu} = \frac{\chi^2}{2N - p}. \quad (3.4)$$

However, χ_{red}^2 should not be interpreted alone, since a low value only reflects the circuit's ability to reproduce the measured spectrum numerically, but not the physical correctness of the ECM [15, 21].

3.3.5 Residual analysis

Equivalent-circuit fitting is a model-dependent mathematical optimisation, where χ_{red}^2 alone is insufficient for rigorous interpretation [15, 17]. It must be complemented by residual analysis (along with the aforementioned lin-KK and parameters uncertainties validation procedures). The fitting process could indeed converge to a mathematically satisfactory fit (low χ_{red}^2 value) with an ECM that does not capture all physical phenomena (underfitting). In order to review the relevance of the fit across the whole frequency range, the complex and relative residuals were defined as:

$$\underline{r}(\omega) = \underline{Z}_{exp}(\omega) - \underline{Z}_{fit}(\omega) \quad \text{and} \quad r_{rel}(\omega) = \frac{\underline{Z}_{exp}(\omega) - \underline{Z}_{fit}(\omega)}{|\underline{Z}_{exp}(\omega)|} \quad (3.5)$$

Fits may be visually satisfactory (Figure 3.2a and Figure 3.2c), however the residual plots are required to highlight any systematic errors in the fitting. For a meaningful fit, the residual plots should show random scatter around zero (Figure 3.2b). Clear periodic patterns across a wide frequency range (Figure 3.2d) usually indicate a poorer overall fit, the ECM used inducing either over or under-fitting [24].

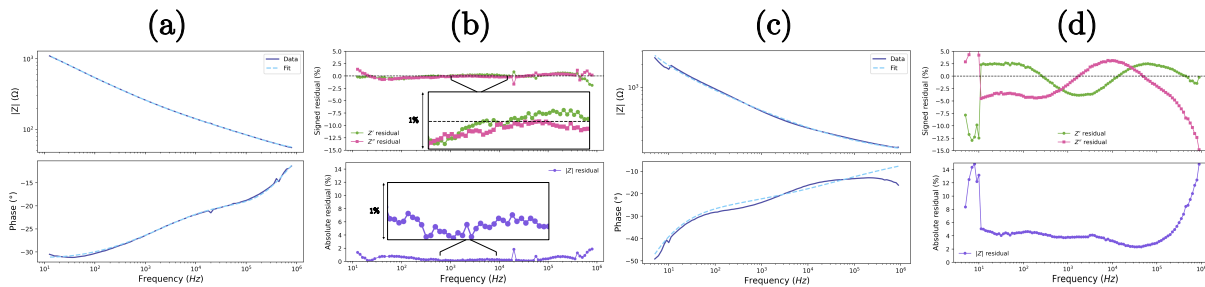


Figure 3.2: Bode plots of different datasets with visually satisfactory fits, but exhibiting really different residuals behaviours: (a) giving small and randomly scattered residuals (b) while (c) gives larger residuals, exhibiting clear patterns (d)

4 Conclusions and outlooks

4.1 Summary of the project

This project developed an automated and configurable Python workflow for the analysis of electrochemical impedance spectroscopy (EIS) data applied to conductive PEDOT:PSS–UPy hydrogels. The scientific motivation was to study soft, hydrated, mixed ionic–electronic materials for bioelectronic and neural-repair applications, where both conductivity and tissue-like properties are important. The methodological motivation was to make EIS analysis more reproducible and scalable, especially for formulation studies where many samples must be compared.

The workflow takes raw impedance spectra and applies a sequence of processing, validation, fitting, analysis, and plotting steps. These include optional outlier removal, frequency-window selection, Link-KK validation, equivalent-circuit fitting, residual analysis, χ^2 and χ^2_{red} calculation, fitted-parameter uncertainty filtering, conductivity calculation, and automated figure generation with user-defined styles.

Experimentally, the workflow was applied to PEDOT:PSS–UPy hydrogel series in which either PEDOT:PSS or UPy content changed. The results support the usefulness of EIS compared with a DC-only conductivity measurement as EIS captures frequency-dependent phenomena which cannot be measured in single DC four-point probes measurements hence allowing for a better characterisation and understanding of the material’s potential applications. More broadly, the project shows that automated EIS analysis can make conductivity trends easier to compare while preserving the link between each fitted parameter, the original spectrum, the chosen circuit, and the validation criteria.

4.2 Limitations

The main limitation in applying EIS to such complex systems is that the measured impedance is a macroscopic, frequency-dependent response rather than a direct observation of the underlying microscopic processes. As illustrated by the blind men and the elephant analogy in Figure 4.1, impedance spectroscopy can reveal different aspects of the system, but the fitted model does not uniquely identify them [15]. A resistance assigned to ionic or electronic transport, or a CPE assigned to double-layer behaviour, may also include overlapping bulk, interfacial, contact, or polarisation contributions. Additional measurements are therefore needed to support the physical interpretation of the equivalent circuit.



Figure 4.1: Spencer Collection, The New York Public Library. "Blind men with an elephant" New York Public Library Digital Collections. Accessed July 10, 2026.

4.3 Future Work

Future work should strengthen both validation and automation. On the EIS side, measurements at several perturbation amplitudes combined with Lissajous-plot inspection would help detect non-linearity and non-stationarity before fitting, while systematic repeated sweeps would quantify drift, swelling, drying, and contact changes during acquisition [30]. Complementary DC or four-point probe measurements should also be used to independently validate conductivities extracted from equivalent-circuit models [6], especially in the two-electrode geometry where hydrogel, contact, and electrode-interface contributions are measured together [15].

Future material libraries should explore a broader formulation space, including PEDOT:PSS content, UPy content, ionic strength, swelling medium, pH, dopants, PEDOT:PSS alternatives, and cytocompatible additives. Biological compatibility should remain central to this screening, since PSS acidity and formulation additives can affect cell viability [26]. Strategies such as modifying the counterion, introducing biocompatible dopants, or using amino-acid or surfactant-based additives such as phenylalanine or DBSA-derived systems may help tune conductivity and cytocompatibility [31, 32]. The resulting datasets could be represented as response surfaces or accuracy contour plots combining conductivity, fit reliability, softness, swelling, stability, and biological response.

Future computational development should also improve model selection and parameter attribution. Instead of only assigning physical meanings to predefined circuits, high-throughput characterisation could approach the circuit-attribution problem (Section 2.1.1) in the reverse direction: systematic changes in formulation could reveal which fitted parameters respond to specific material features. This would help determine whether fitted resistances are linked to ionic transport, electronic transport, interfacial polarisation, or mixed contributions.

Overall, the workflow developed here represents a first step toward a closed-loop materials-discovery platform, where formulation, characterisation, analysis, and optimisation are connected through automated measurement and data-driven decision-making, as illustrated in Figure 4.2.

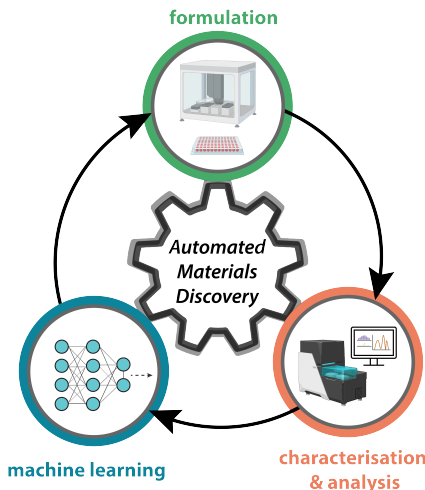


Figure 4.2: Conceptual illustration of a closed-loop materials-discovery workflow. The present project contributes mainly to the characterisation and analysis step by automating EIS data processing, validation, fitting, and conductivity extraction.

5 Appendices

5.1 Introduction

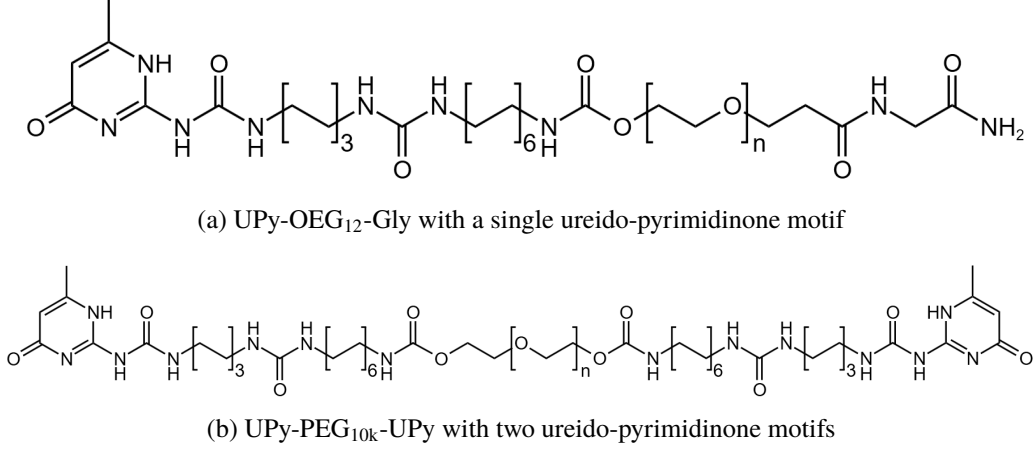


Figure 5.1: Chemical structures of the UPy derivatives used in this work: (a) monofunctional UPy-OEG12-Gly and (b) bifunctional UPy-PEG10k-UPy. Additional details on UPy network formation can be found in Rijns [11, 12, 13].

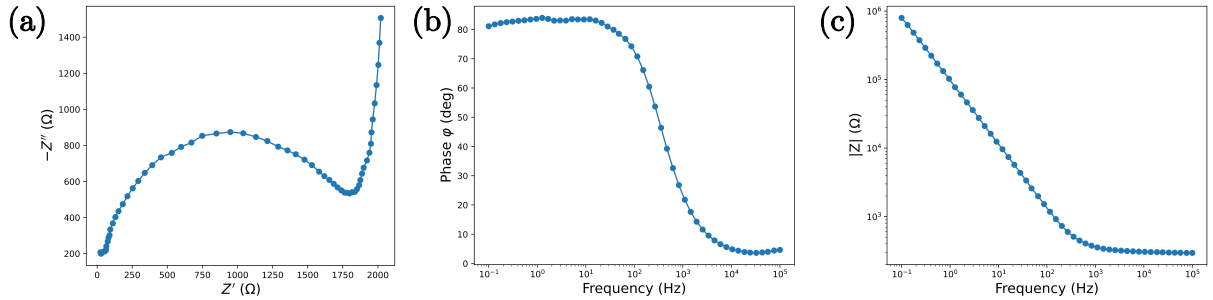


Figure 5.2: (a) Nyquist plot of PEDOT:PSS-UPy Hydrogel in dilute state with setup Figure 3.1b (b), (c) Phase φ and modulus $|Z|$ Bode plot for frequency range 0.1Hz to 1MHz

5.2 Results

5.2.1 Workflow

The actual total impedance of the circuit shown in Figure 2.1a is as follows:

$$\underline{Z}_{tot}(\omega) = \frac{R_e(1 + R_i Q(j\omega)^\alpha)}{1 + (R_e + R_i) Q(j\omega)^\alpha} \quad (5.1)$$

Theoretically, it should thus be observed that $\underline{Z}(\omega) \underset{\omega \rightarrow 0}{\sim} R_e$ and $Z_{tot}(\omega) \underset{\omega \rightarrow \infty}{\sim} \frac{R_e R_i}{R_e + R_i}$. However, due to long measurement time and non-stationarity [16, 17], it is not possible to measure low enough frequencies to have $Z_{tot} \approx R_e$, it is indeed observed that, at the lowest measured frequencies, $\varphi \approx \frac{\pi}{2}$ while one would expect $\varphi \approx 0$ since $R_e \in \mathbb{R}$. This is why Figure 2.1.a should be interpreted as a conceptual guide rather than as a complete physical description of the measured spectra.



Figure 5.3: Usual file structure when using the EIS_fitting library, the raw data (a), the library (b), and the processed data (c) are kept in three different folders.

```
{
  "paths": {
    "data_path": "/home/enzos/data/",
    "user": "femke",
    "project": "article",
    "output_mode": "workspace",
    "output_path": "/home/enzos/data_processed"
  },
  "run": {
    "pre_plot": false,
    "plot": true,
    "compute_mean_parameters": false
  },
  "outlier_removal": {
    "enabled": true,
    "threshold": 1.5,
    "features": [
      "log_modulus",
      "phase"
    ]
  },
  "validation": {
    "display_chi_square": false,
    "fitting_frequency_range": [
      10,
      null
    ],
    "plot_residuals": true,
    "lin_kk": {
      "enabled": true,
      "reject_on_fail": true,
      "c": 0.85,
      "max_M": 50,
      "fit_type": "complex",
      "add_cap": true
    },
    "error_percent_param_threshold": 100
  },
  "plotting": {
    "R_and_Q": {
      "left_ylim": [
        1e-1,
        1e15
      ],
      "right_ylim": [
        1e-10,
        1e-1
      ]
    },
    "conductivities": {
      "left_ylim": [
        null,
        null
      ],
      "normalize": true,
      "cmap_name": "Oranges",
      "log_scale": false,
      "legend": {
        "rotation": 0,
        "ha": "center",
        "fontsize": 14
      }
    }
  },
  "residuals": {
    "ylim_signed": [
      null,
      null
    ],
    "ylim_absolute": [
      null,
      null
    ]
  },
  "save_preferences": {
    "transparent": true,
    "png": true,
    "svg": false,
    "dpi": 400,
    "full_terminal": true
  }
}
```

Figure 5.4: Example of a `global.json` config file in this workflow

When relevant to the experiment design, the option `compute_mean_parameters` in `global.json` (Figure 5.4) can be enabled to add mean parameter values, computed with the formula defined in Equation (5.2), directly to the circuit parameter CSV files.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad u(x) = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (5.2)$$

These mean values $\bar{x}$ and their respective through-sample uncertainties $u(x)$ are useful for reporting repeatability of results over frequency sweeps and samples. Only successful and accepted fits should be used for such averages.

Remark: Some studies report an effective capacitance calculated from CPE parameters, which can be useful for comparing non-ideal capacitive responses between samples. However, this conversion is model-dependent: Q is not a true capacitance unless $\alpha = 1$, and different formulae are required depending on the circuit topology and on whether the CPE represents a surface distribution, a normal distribution, or another physical origin. For this reason, effective capacitances were not treated here as primary physical observables. When used, they were interpreted only as comparative descriptors and always together with Q , α , the fitted circuit topology, and the residuals [15, 33, 34].

5.2.2 Experimental Results

Fixed UPy w/v%

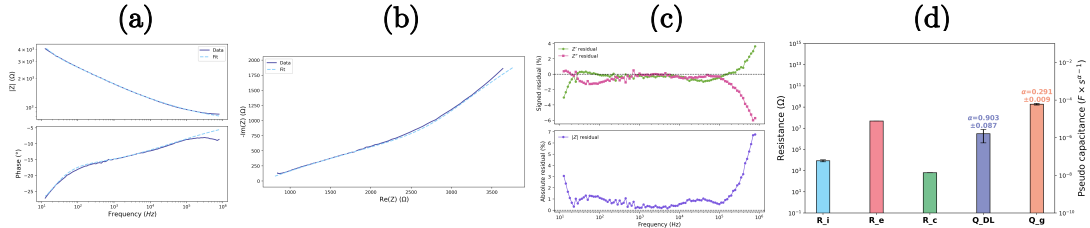


Figure 5.5: Example of the automated fitting and analysis outputs generated by the workflow in the case of 0.1 w/v% PEDOT:PSS. (a) Bode representation comparing the experimental impedance spectrum with the fitted equivalent-circuit response obtained with `circuit_2` defined in Figure 2.3. (b) Nyquist representation of the same fit. (c) Residual analysis showing the signed real and imaginary residuals, together with the absolute residual as a function of frequency. (d) Extracted circuit parameters, with resistance elements plotted on the left axis and CPE pseudo-capacitance elements on the right axis.

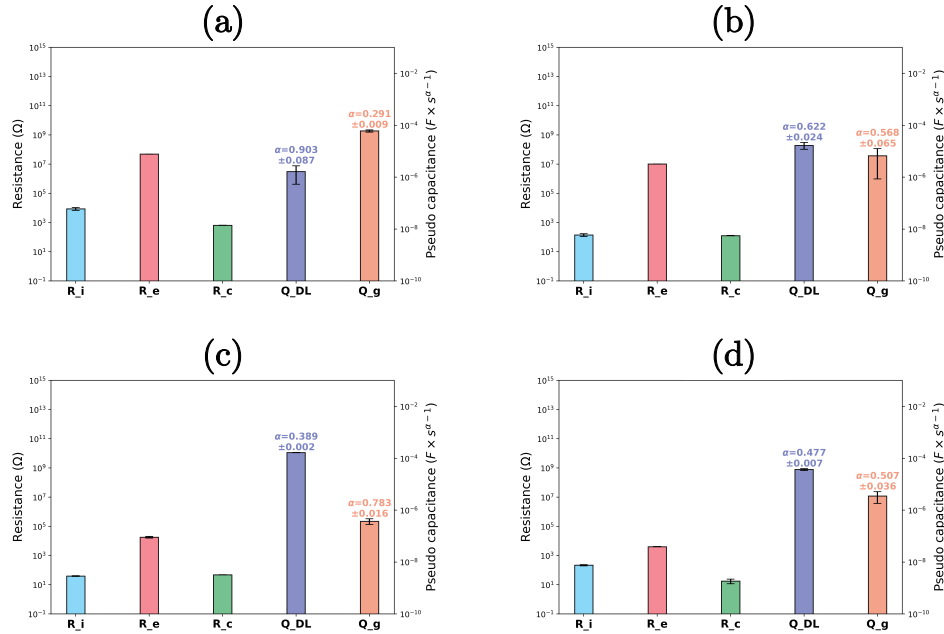


Figure 5.6: Fitted circuit parameters obtained for the PEDOT:PSS with `circuit_2` (Figure 2.3) concentration series at fixed UPy content with PEDOT:PSS w/v% of respectively 0.1, 0.5, 1.0, and 2.0 from (a) to (d). Resistance parameters are plotted on the left axis, while CPE pseudo-capacitance parameters are plotted on the right axis. The associated α values are indicated above the corresponding CPE bars. Error bars represent the fitted standard deviations.

Fixed PEDOT:PSS w/v%

Because R_1 and Q_1 are generally smaller than R_2 and Q_2 , the first parallel R -CPE block is expected to have the shorter characteristic time and is therefore labelled as a high-frequency contribution. The second block is labelled as a lower-frequency contribution, while the terminal CPE_{ep} accounts for the low-frequency electrode polarisation tail. These labels are descriptive rather than definitive physical assignments, since both relaxations may contain overlapping bulk, interfacial, ionic, and electronic contributions.

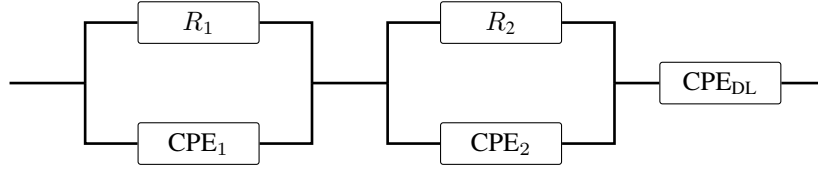


Figure 5.7: Equivalent circuit model `circuit_3`, inspired by a circuit used by Orazem et al. to describe systems with two distributed relaxation processes [15, 16].

Table 5.1: Fitted parameters extracted for circuit 3 across the three formulations. The uncertainties correspond to the fitted standard deviations. Resistances are given in Ohms (Ω) and pseudo-capacitances Q_i in $F \times s^{\alpha_i-1}$. For the product $R_k Q_k$, the uncertainty was propagated from R_k and Q_k .

Sample	$\chi^2_{\text{red}}/10^{-4}$	R_1	$Q_1/10^{-7}$	α_1	$R_1 Q_1/10^{-4}$
P2U0.6	1.918	260.4 ± 6.5	3.650 ± 0.839	0.5845 ± 0.0143	0.9503 ± 0.2198
P2U1.2	4.386	196.9 ± 7.8	85.22 ± 25.21	0.4159 ± 0.0179	16.78 ± 5.01
P2U2.5	2.647	73.82 ± 4.34	17.63 ± 7.22	0.5608 ± 0.0266	1.301 ± 0.539

Sample	$R_2 \times 10^2$	$Q_2 \times 10^{-5}$	α_2	$R_2 Q_2 \times 10^{-2}$
P2U0.6	34.05 ± 1.24	1.462 ± 0.065	0.5029 ± 0.0059	4.978 ± 0.286
P2U1.2	13.56 ± 1.52	2.627 ± 0.201	0.5752 ± 0.0153	3.562 ± 0.483
P2U2.5	9.113 ± 0.454	1.661 ± 0.109	0.7471 ± 0.0143	1.514 ± 0.124

Section 4.3 discusses how future work could improve the attribution of each parameter to specific physical phenomena.

Remark: The two R -CPE blocks have the same mathematical form and are therefore interchangeable in the circuit topology. As a result, the assignment of the first block to a high-frequency contribution and the second block to a low-frequency contribution depends solely on the order of the elements in the circuit definition and on the initial guesses provided before fitting. These labels are therefore descriptive fitting labels rather than unique physical assignments.

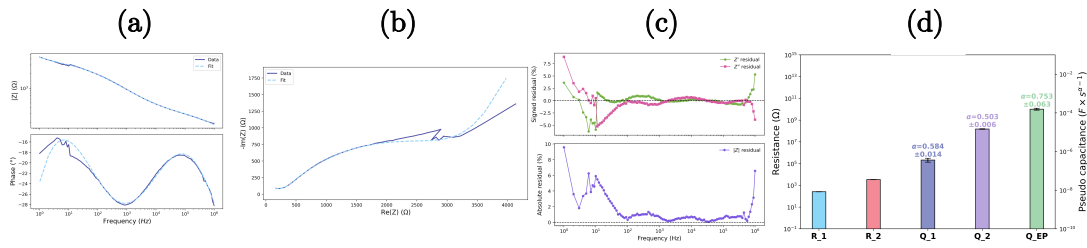


Figure 5.8: Example of the automated fitting and analysis outputs generated for the P2U0.6 hydrogel using `circuit_3`. Bode and Nyquist representations (a), (b) Nyquist. (d) Extracted circuit parameters, with resistance elements plotted on the left axis and CPE pseudo-capacitance elements plotted on the right axis. (c) residuals plotting.

5.3 Materials and Methods

Table 5.2: EIS measurement parameters used for the frequency sweep. Note that Number of repeats is the number of repetitions **after** the first sweep, hence the aforementioned “three sweeps per sample” in Section 3.2. No external DC bias was intentionally applied during the EIS measurements.

Parameter	Symbol	Value
Initial frequency	f_i	1.000 MHz
Final frequency	f_f	1.000 Hz
Points per decade	N_d	20
Frequency spacing	–	Logarithmic
Sinusoidal amplitude	V_a	10.0 mV
RMS amplitude	V_{rms}	≈ 7.07 mV
Waiting time before each frequency	p_w	0.15 period
Averages per frequency	N_a	4 measurements
Drift correction	–	Disabled
Number of repeats	n_c	2

Table 5.3: Measurements of the experimental setup used to obtain the spectra presented in this report, made with a numerical calliper

Parameter	Value
Platinum plate thickness	3.20 mm
Well depth	10.73 mm
Well diameter	6.13 mm
Inter-electrode distance	6.99 mm

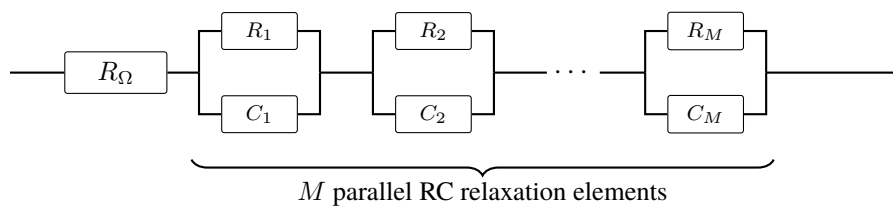


Figure 5.9: Schematic representation of the auxiliary Lin-KK circuit used for impedance validation

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